

ZANDER HIRMAN

Boulder, CO | 303-834-2108 | zhirm08@outlook.com | [linkedin.com/in/zander-hirman](https://www.linkedin.com/in/zander-hirman)

Geography graduate with field mapping experience across government, conservation, and international contexts — including shoreline analysis for the National Park Service, seed-collection mapping for the Colorado Native Plant Society (140+ field hours), and satellite imagery digitization for DRCOG. Comfortable working independently in the field and delivering clean, QC'd spatial data.

EDUCATION

University of Colorado Boulder

Boulder, CO

Bachelor of Arts in Geography | GPA: 3.32 | May 2026

Relevant coursework: Machine Learning & Spatial Data, Statistics & Geographic Data, Environment-Society Geography, Political Ecology of Latin America, Interdisciplinary Data Science

EXPERIENCE

Independent GIS Field Mapping Project — Chipanga Village, Njombe Region, Tanzania

2026

Self-directed field data collection project supporting a community water & infrastructure initiative

- Collected and geolocated 100+ water and infrastructure features in the field over 3 days using QGIS and QField, spanning a 7-layer dataset (water towers, pipelines, solar arrays, fencing, garden plots, school buildings)
- Designed a compact field-coding system for QField's simplified Notes interface, enabling efficient attribute capture in a low-connectivity environment and accurate later transcription
- Calculated field-based water usage estimates and delivered final maps and a project website used to communicate findings to investors and stakeholders

Colorado Native Plant Society (CoNPS) — Field Research Intern

Fall 2025

- Logged 140+ field hours traveling to sites across Colorado to collect native plant seeds as part of a statewide conservation effort
- Built and published an interactive web map in ArcGIS Online, integrating it into a public-facing StoryMap detailing CoNPS's history, mission, and seed-collection regions ([view StoryMap](#))
- Mapped and documented the geographic extent of collection plots and regions used by the organization

Golden Gate National Recreation Area (GOGA), National Park Service

Spring 2022

- Mapped shoreline sites to monitor and track starfish population distribution along the coast
- Analyzed shoreline change over time, extending the study area south to Point Reyes

Denver Regional Council of Governments (DRCOG)

Winter 2021

- Digitized 1,000+ land parcels, building rooftops, and fence lines from satellite imagery to support regional planning datasets
- Performed quality control review on digitized features to ensure spatial accuracy across the dataset

ACADEMIC PROJECTS

Predicting Wildfire Risk in Colorado Using Spatial Embeddings & Machine Learning

2025

Graduate-level Machine Learning course project

- Built a wildfire ignition risk model designed to help identify at-risk wildland-urban interface areas for preparedness and resource planning, using Google's AlphaEarth satellite embeddings combined with environmental covariates (NDVI, climate, terrain, fuel data) across Colorado, New Mexico, Wyoming, and Utah
- Trained and evaluated 5 models (Logistic Regression, Random Forest, XGBoost, LightGBM, Neural Network); best-performing XGBoost model achieved 0.86 AUC and 0.83 recall after threshold tuning, prioritizing recall to minimize missed fire risk
- Sourced and integrated data from 8 independent sources including Sentinel-2, ERA5, USGS SRTM, LANDFIRE, NLCD, and NOAA VIIRS

Migration Route Mapping: Mexico to the United States

Spring 2025

- Mapped and visualized a migration route from Mexico to the U.S., inspired by the memoir *Solito*, integrating geographic and narrative data into a single spatial story

CERTIFICATIONS

Esri ArcGIS Technical Associate: In Progress

SKILLS

GIS & Mapping: QGIS, QField, ArcGIS Pro, ArcGIS Online, StoryMaps

Remote Sensing: Google Earth Engine, AlphaEarth embeddings, Sentinel/Landsat imagery analysis

Programming & Data: Python, R, SQL, Machine Learning (scikit-learn, XGBoost, LightGBM), Git/GitHub

Web Mapping: HTML, CSS, JavaScript, Leaflet, Mapbox

Other: Microsoft Office Suite, Google Workspace